IBM Research

Hybrid Quantum-Classical Graph Representation Learning

Quantum and Classical Walks

Continuous-time Random Walk

Algorithm Variations

Classical & Quantum

Classical

$$\mathbf{P}(t) = e^{-t\mathbf{M}}$$

$$p_{uv}(t) = \mathbf{P}(t)[u, v]$$

Quantum

$$\mathbf{Q}(t) = e^{-it\mathbf{H}}$$

$$p_{uv}(t) = |\mathbf{Q}(t)[u, v]|^2$$

Discrete-time Random Walk

Algorithm Variations

Classical & Quantum

Classical

$$p_{j \to k} = P(X_{n+1} = k \,|X_n = j)$$

$$X_n = \sum_{k=0}^{n-1} \eta_k$$

Steps taking $+1$ or -1 with $p = \frac{1}{2}$

Quantum

$$p_{j \to k}(t) = \sum_{k=1}^{d_j} |\psi_{j,k}(t)|^2$$

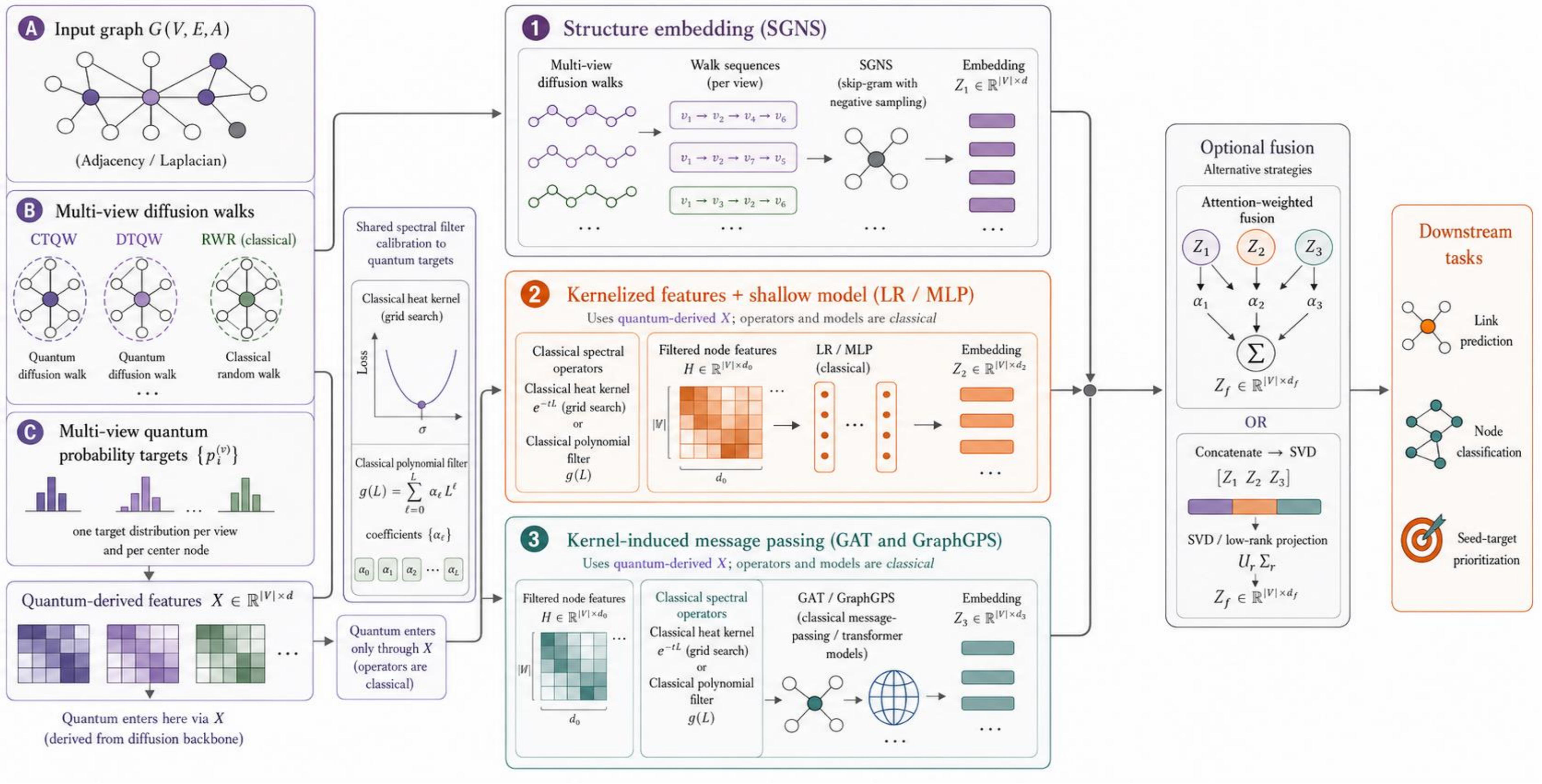
$$|\psi(t)\rangle = U^t|\psi(0)\rangle$$

$$U = SC$$

Shift operator

Coin operator

A framework for Quantum-enabled Graph Representation Learning



Graph Complexity Measures

We study graph complexity measures to understand which complexity measures are drivers of the quantum vs. classical graph representation learning for downstream tasks.

Structural Topology

- $\log(\#\ of\ nodes)$
-
- Density
- Average Degree
- Modularity
- Largest CC fraction
- Cycle Density
- …

Spectral Structure

- Spectral Gap
-
- IPR
- Spectral Entropy
- Log Odd Girth
- Algebraic Connectivity
- …

Diffusion

- Average Path Length
- Heat Kernel Trace
- Approx. Conductance
- Non-backtracking radius
- …

Transport Geometry

- Oliver-Ricci Conjecture
- Pagerank Gini
- Betweenness Gini
- Closeness Gini
- …

IBM Research

Graph Complexity Measures

Structural topology

Degree heterogeneity (Gini)

Homogeneous degrees → weak hub interference

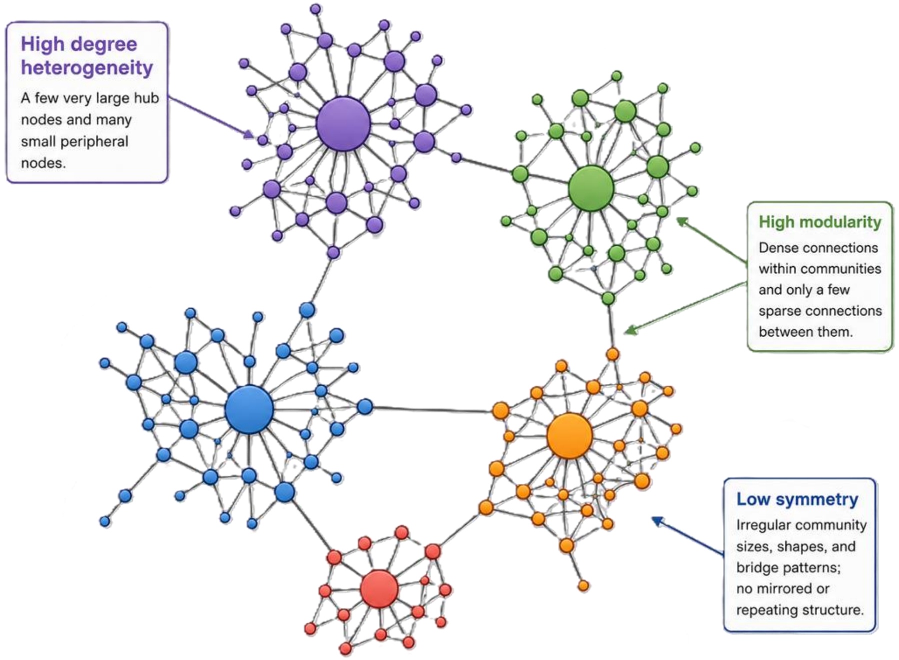
Modularity / community

Few bottlenecks for QW to exploit

Symmetry (WL compression)

More automorphism → classical-tractable

*Quantum-favoring regime**



low high

Desired value

Classical-favoring regime: Homogeneous, weakly-clustered, high-symmetry graphs.

* Probable regime for quantum walks outperforming classical walks

IBM Research

Graph Complexity Measures

Spectral Structure

Eigenvector Localization (IPR)

High IPR → QW collapses to classical diffusion

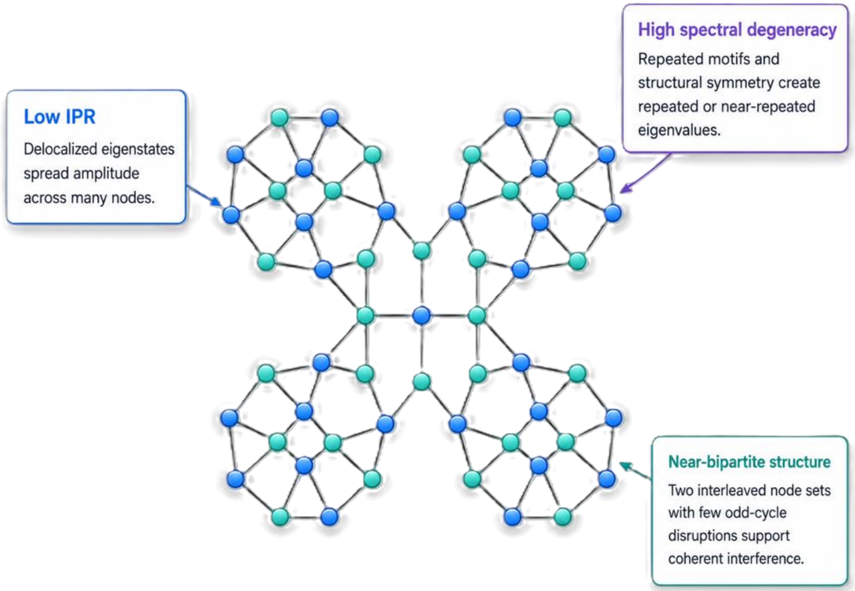
Spectral Degeneracy

Few degeneracies → weaker coherent interference

Bipartite Proximity

Far from bipartiteness → no DTQW channel

*Quantum-favoring regime**



Classical-favoring regime: localized eigenstates, non-degenerate spectrum, far from bipartite.

Note bipartite_proximity = $\max(0, 2-\lambda_{max})$*: 0 means bipartite, so 'far from bipartite' is a HIGH metric value — arrow points high to match the code's sign convention.*

* Probable regime for quantum walks outperforming classical walks

Graph Complexity Measures

Diffusion & Mixing

Spectral Gap

Fast classical mixing → little room for QW gain

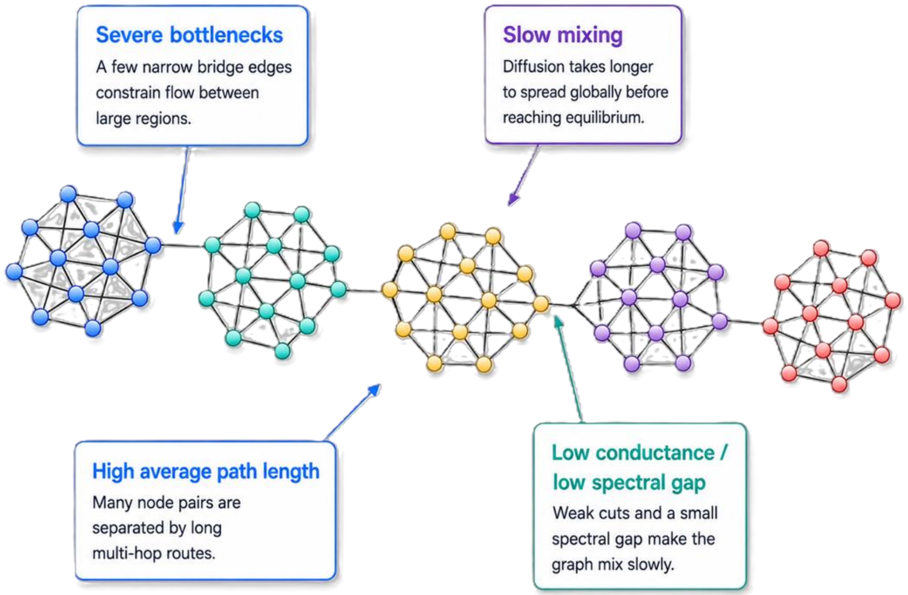
Conductance

No tight bottleneck → classical transport adequate

Average Path Length

Short paths → classical reach already efficient

*Quantum-favoring regime**



low high

Desired value

Classical-favoring regime: well-connected, fast-mixing graphs with no severe bottleneck.

* Probable regime for quantum walks outperforming classical walks

Graph Complexity Measures

Transport Geometry

Negative ORC fraction

Fewer negatively-curved bottleneck edges

Betweenness Gini

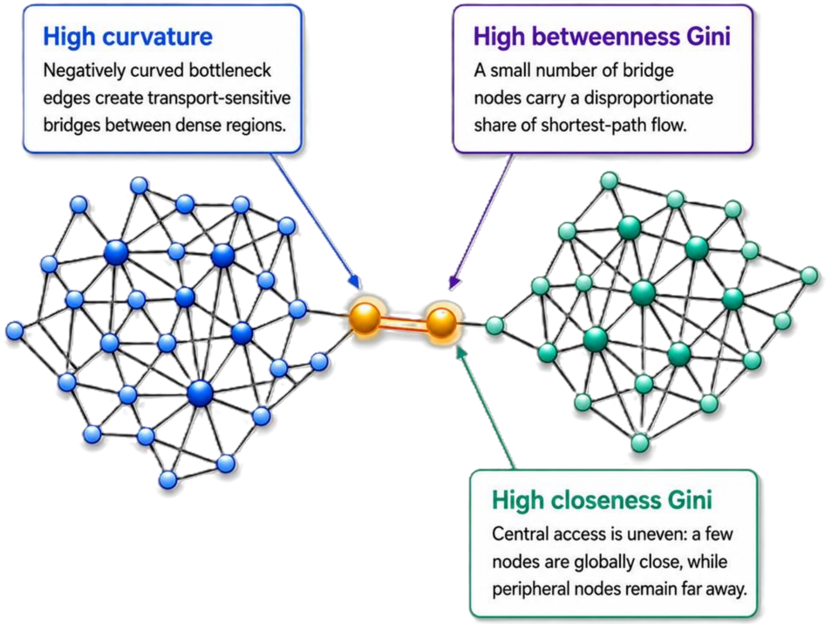
Even load → no congestion for QW to bypass

Closeness Gini

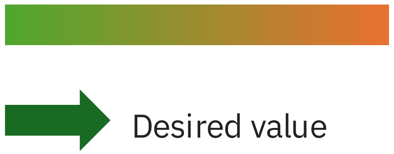
Uniform reach → classical walk suffices

Classical-favoring regime: near-flat curvatures and even load distribution.

*Quantum-favoring regime**



low high



Desired value

* Probable regime for quantum walks outperforming classical walks

Benchmarking with QuVINE

- We applied the QuVINE framework to over 70 networks encompassing synthetic and real-world data with three different sizes (500, 2000, 5000).

- For each network type and node size, we studied over 20 graph complexity metrics and evaluated three tasks: node classification, link prediction, and node prioritization/ranking.

- We designed a comprehensive walk-based benchmarking suite where we generated network embeddings using quantum and classical walks in four distinct training strategies:
  - ❖ Skip-Gram Negative Sampling (Word2vec)
  - ❖ Walk-calibrated diffusion heat and polynomial kernels
  - ❖ Graph Attention Networks
  - ❖ General, Powerful, Scalable Graph Transformer (GraphGPS)

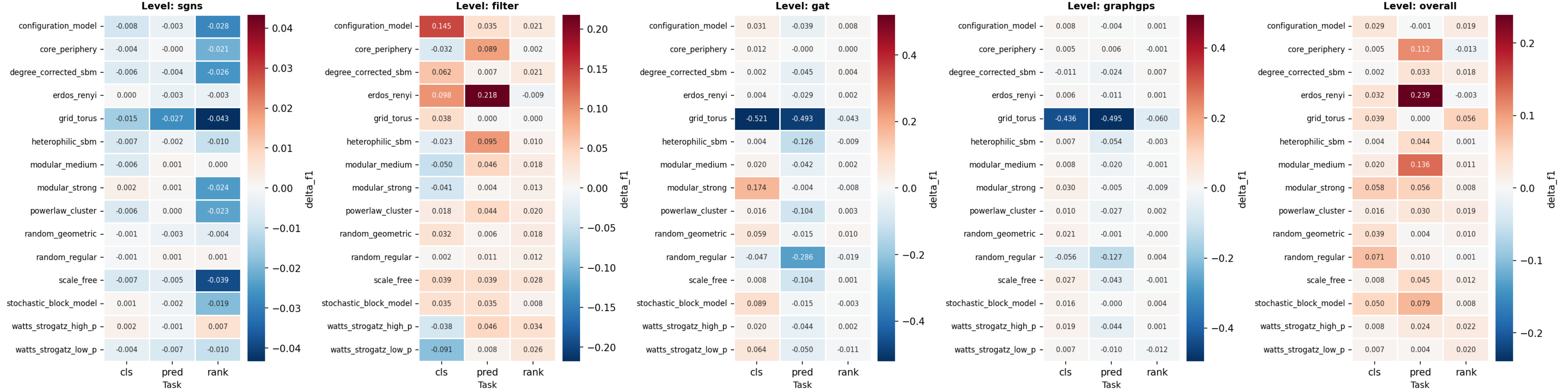
- For each, we used continuous- and discrete-time quantum walks (CTQW and DTQW) and random walk with restart (RWR)-based calibration yielding a total of 18 quantum and 17 classical embeddings.

- On top of this, as classical state-of-the-art, we used five methods from different families of network embedding algorithms, namely, Node2vec, NetMF, GraphSAGE, Graph Convolutional Network, and APPNP.

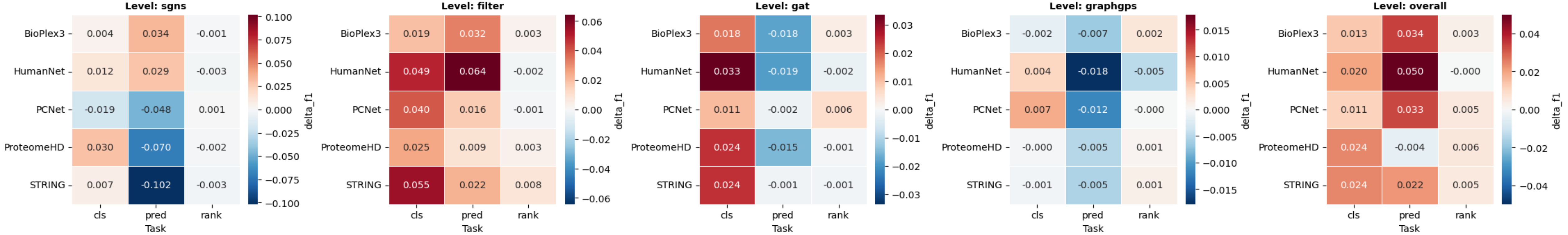
- For each method, we ran 30 different test-train splits for each task, yielding $70 \times 3 \times 3 \times 30 \times 40 = 756{,}000$ experiments and an additional ~ 4000 experiments for tuning hyperparameters resulting in $\sim 760{,}000$ experiments.

Benchmarking Results

Summary: mean delta_f1 by Network Type × Task (synthetic networks)
(positive = classical wins, negative = quantum wins)
[cls=mean_f1_macro, pred=inner_product, rank=p@10_centroid]



Summary: mean delta_f1 by PPI network × task
(positive = classical wins, negative = quantum wins)



Benchmarking Results